# NANDITA KUMARI

Website |Google Scholar | Github | Contact

## EDUCATION

- **PhD in Geosciences**, Stony Brook University GPA **3.96**/4.0 [Fall'19-Present]

- **B. E in Geoinformatics**, College of Engineering Guindy, Anna University, Chennai, GPA **8.93**/10 [2018 ]

## HONOURS

- **NASA Exploration Science Forum** Travel Award, 2023
- **Zonta International Amelia Earhart Fellowship,** 2023
- **Featured Early Career Presentation**, Lunar Exploration Analysis Group (LEAG) Meeting 2022
- **LEAG Travel Award**, 2022
- **Best Poster Award** in National Space Science Symposium 2019
- **Mike Barnsley Memorial Award**[1] by Alumni Association of CEG
- **Alumni Association Endowment Award 2007-2**[2] in Geoinformatics Engineering batch 2018
- **Merit Scholarship** from Army Welfare Association

## INVITED TALKS

**VELAY MEETING,** Stony Brook University, 2023

## PROFESSIONAL EXPERIENCE AND SERVICE

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|---|---|---|
| **CHAIR** | *NASA EXPLORATION SCIENCE FORUM 2020, 2023* | |
| **REVIEWER** | *ICARUS, PSJ, GRL* | |
| **EXTERNAL REVIEWER** | *NASA PANEL 2023* | |
| **EXECUTIVE SECRETARY** | *NASA PANEL 2022* | |
| **MENTOR** | *GWISE* | [FALL'22-SPRING'23] |
| **CHAIR** | *LUNAR EXPLORATION ANALYSIS GROUP MEETING 2022* | |
| **SUMMER INTERN** | *LUNAR AND PLANETARY INSTITUTE* | [MAY'20-AUG'20] |
| **VISITING STUDENT** | *PHYSICAL RESEARCH LABORATORY* | [MAY'18-JUL'19] |
| **SUMMER INTERN** | *PHYSICAL RESEARCH LABORATORY* | [MAY'16-JUN'16] |

## PUBLICATIONS & ABSTRACTS

1. SURFACE CONDITIONS AND RESOURCE ACCESSIBILITY AT POTENTIAL ARTEMIS LANDING SITES 007 AND 011 ( *https://iopscience.iop.org/article/10.3847/PSJ/ac88c2)*
2. GEOLOGICAL CHARACTERIZATION OF CHANDRAYAAN-2 LANDING SITE IN THE SOUTH POLE REGION OF THE MOON *(https://doi.org/10.1016/j.icarus.2019.113449)* [CO-AUTHORED]
3. THE SCIENTIFIC VALUE OF A SUSTAINED EXPLORATION PROGRAM AT THE ARISTARCHUS PLATEAU (*https://iopscience.iop.org/article/10.3847/PSJ/abfec6 )* [CO-AUTHORED]
4. TEMPERATURES OF THE LACUS MORTIS REGION OF THE MOON (*https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2021EA001966* ) [CO-AUTHORED]
5. SPACE WEATHERING ACROSS THE SOLAR SYSTEM: LESSONS FROM THE MOON AND OUTSTANDING QUESTIONS *(https://baas.aas.org/pub/2021n4i172/release/1).* [CO-AUTHORED]
6. MISSION TO CHARACTERIZE VOLATILES IN OLD, COLD, PERMANENTLY SHADOWED REGIONS ON THE MOON (*https://baas.aas.org/pub/2021n4i365/release/1 )* [CO-AUTHORED]

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[1] Annual award for highest marks in image processing in theory and lab

[2] Department rank 1 for the academic year 2016-2017

7. Different Trends of Variation in Christiansen Feature on the Lunar Surface : Effects of Iron Content and Particle Size [In Review, JGR]
8. Bulk Composition Characterization of The Lunar Surface Using Machine Learning: Applications to Ongoing and Upcoming Missions [NESF 2023]
9. Brewing Machine Learning Models For Accurate Analysis of Current and Future Datasets [LPSC 2023]
10. Procedure Standardization For Toolbox for Research and Exploration (TREX) Field Data Analysis [LPSC 2023]
11. Gruithuisen Domes: Silicic Volcanic Constructs on The Moon [LEAG 2022]
12. A Holistic View of Gruithuisen Domes – Silicic Constructs on the Moon [LPSC 2022]
13. Surface Illumination, Temperatures and Cold Traps at Two Potential Landing Sites Near the Lunar South Pole [NESF 2021]
14. Mid-Infrared Spectral Characteristics of Dust (Minerals and Meteorites) and Their Utility for Interpreting Remote-Sensing Data.[SOFIA Rock, dust and Ice Workshop 2021]
15. Episodic Volcanism and Fluvial Activities in a Floor-Fractured Crater at the Flanks of Arsia Mons [LPSC 2021]
16. Geological Diversity at Two Potential Landing Sites in the Lunar South Pole[LPSC 2021]
17. Artemis III EVA Opportunities on the Rim of de Gerlache Crater [NASA requested input for Artemis III Science Definition Team. Co-Authored]
18. Alternate Artemis III EVA Opportunities near de Gerlache Crater [NASA requested input for Artemis III Science Definition Team, Co-Authored]
19. Artemis III EVA Opportunities on the Lunar Farside near Shackleton Crater [NASA requested input for Artemis III Science Definition Team, Co-Authored]
20. Different Trends of Variation in Christiansen Feature on The Lunar Surface: Effects of Iron Content and Particle Size [NESF 2020] [TALK]
21. Investigation of Variation in Christiansen Feature With Respect To Albedo on the Moon Using the Diviner, $M^3$ and Kaguya Datasets [LPSC 2020] [PSWG TALK]
22. Global Distribution and Characteristics of Domical Craters on the Moon [LPSC 2019]
23. Complex Impact Craters within Lunar Basins and Their Implication [NSSS 2019]

## Field Work

Kilbourne and Hunts Hole, SSERVI (RISE4) 2022;2023

San Francisco Volcanic Field, SSERVI (CLSE) 2022

Utah Yellowcat SIte, SSERVI(TREX),2022

Mount Ascutny, *New England NEIGC Field Trip 2019*

## Leadership Skills

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|---|---|---|
| **Senator** | \| Graduate Student Organization of SBU | [2021-Present] |
| **Student Director** | \| *CEG Tech Forum* | [2014-2018] |
| **Director** | \| *Literary Club of Anna University* | [2018] |
| **Core Member** | \| *Society for Geoinformatics Engineers* | [2014-2018] |
| **Experience Manager** | \| *AIESEC International, Chennai* | [Jul'15– Jul'16] |

## Technical Skills

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| **Programming** | Python(Data Analysis/Machine Learning, Proficient) |
| **Software** | ArcGIS, QGIS, ENVI, ERDAS Imagine, ISIS, DaVinci |

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| **DATA PROFICIENCY** | Earth: LANDSAT[3] (VIS, THERMAL, PAN), ASTER Thermal InfraRed Data; Moon: LRO (WAC, NAC, WAC-DTM, Diviner, LOLA), Chandrayaan-1($M^3$), SELENE, KAGUYA ; Mars: MRO (CRISM, CTX, HiRISE), MGS(MOLA) |

[3] LANDSAT Data from 5 to 8